

Pranav Kumar Soma

SOFTWARE ENGINEER

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San Francisco Bay Area, CA • Available January 2027

EDUCATION

M.S. Computer Science, University of California San Diego

GPA 4.00/4.00 • Jacobs School of Engineering • completed in 1 year

Jan 2026 – Dec 2026

La Jolla, CA

B.S. Computer Science, University of California San Diego

GPA 3.96/4.00 • Cum Laude, CSE Department Honors, Provost Honors every quarter • completed in 2 years

Sep 2023 – Dec 2025

La Jolla, CA

Coursework: Graduate Algorithms, Operating Systems, Distributed Systems, Computer Security, Theory of Computation, Software Engineering, Principles of Programming Languages, Data Systems for ML, Computer Vision, Statistical NLP.

EXPERIENCE

Software Engineer Intern, ServiceNow

Jun 2026 – Sep 2026

AI Voice Agent Analytics

San Diego, CA

- Shipped a full-stack conversation-history troubleshooting platform unifying **20+ data tables** into overview and per-call drill-down reports; released to production across **24+ enterprise instances** serving **10K+ voice calls/day**
- Cut report load **7s to under 1s** over millions of records (up to 100 interaction records per call) via lazy loading, caching, and an iterative scalability redesign; React, JavaScript, Playwright suites in Jenkins CI/CD
- Owned scope through delivery: authored PRDs and design docs, ran design reviews and sprint demos, presented to a **VP and Senior Director**, and closed high-priority defects and customer escalations solo

Software Engineer Intern, ServiceNow

Jun 2025 – Sep 2025

AI Voice Agents, IT Service Management • **Technical Skills Award**, top intern of 800+

San Diego, CA

- Owned the **AWS Lex V1 to V2** migration end to end (8 intents, 48+ slots) ahead of end-of-life: **45% lower agent-response latency**, shipped to production and validated live with an enterprise customer. **Return offer**
- Built “Wrap Up,” an AI call-summarization flow (Seismic, custom data brokers, JavaScript client scripts) cutting average call-closing time **6 minutes** across **10K calls/day** per customer telemetry

AI Research Fellow & Agentic Systems Lead, Laboratory for Emerging Intelligence, UCSD

Nov 2024 – Present

- Lead **6 engineers** on a **Teradata-funded** agentic workbench (**100+** pilot users): **60+ MCP servers** over all **600 UCSD APIs**, a planner/executor split that runs workflows deterministically and degrades gracefully; FastAPI, MongoDB, GCP Cloud Run
- Built and deployed the Python **AI Tutor** to UCSD, SDSU, and Cal Poly Pomona across 10 courses; a controlled study over **600+ students** measured **64% fewer TA overtime hours**

Engineering Manager, Triton Software Engineering

Sep 2024 – Jun 2025

Software Developer, Sep 2023 – Sep 2024

San Diego, CA

- Led **11 developers and designers** through two delivery cycles of Spay.LA, a MERN fundraising and legislative platform that **raised \$75K**; scoped client requirements into features and owned delivery
- Built the Union Station Homeless Services housing portal in **TypeScript MERN** for LA County’s sole provider of those services: media upload with client-side downsizing, validated intake, state-machine resubmission flows, bulk MongoDB exports

Computer Vision & Embedded Systems Engineer, Triton Robotics

Sep 2023 – Jun 2024

50-person competitive robotics team

San Diego, CA

- Built real-time motor-control pipelines in **C/C++ over ROS** and Python autonomous-movement CV; implemented conditional **pure-pursuit** planning, PID controllers, and contour-detection tracking

Honors & M.S. Thesis Researcher, UCSD CSE, Computational Redistricting

Sep 2024 – Present

- Built 3 multi-threaded districting generators over Census 2020 data at **158K to 5.5M blocks/state** (XGBoost + graph partitioning, Sequential Monte Carlo, greedy DP with Kernighan–Lin); Python, pymetis, GeoPandas, Parquet

SELECTED PROJECTS

- ORCA, AI Music Production** (Best Project, SDTC 2025: 1 of 30 from 200+) • OpenCV score reading, neural source separation, CNN audio-to-score; **98% arrangement accuracy**. Flask, Docker, React
- TSE Constellation** • helped design and build the private early version of Triton SE’s shared React/TypeScript component system (later published as @tritonse/tse-constellation), serving a **100+ member** engineering organization

LEADERSHIP & HONORS

Co-Founder & President, Real World AI Network (2026–Present) • **7,000+ student** federation, **50+ faculty**, 6 schools.

President, CSE Society @ UCSD (2025–2026) • **300+** members, **21+** projects, sponsors **2 to 16**, **\$50K+** raised, first **Linux Foundation** hackathon.

Honors: ServiceNow Technical Skills Award • Best Project SDTC 2025 • Cum Laude & CSE Department Honors • YC Startup School 2026 • National Merit Finalist • USACO Silver.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, TypeScript, JavaScript, SQL, Bash, ARM Assembly, Swift, Kotlin, HTML/CSS

Web & Backend: React, Node.js, Express, FastAPI, Flask, Django REST, MERN, REST APIs, Firebase, Google OAuth; MongoDB, PostgreSQL, MySQL; Playwright, JUnit, Postman, Agile/Scrum, system design, design docs and PRDs

Systems & Cloud: Docker, Kubernetes, Jenkins CI/CD, GCP (Cloud Run), AWS (Lex, Lambda), Git, Linux, ROS; distributed, parallel, and real-time systems, finite state machines; PyTorch, XGBoost, OpenCV, LLM fine-tuning, RAG, LLM agents and MCP